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P6-A · Transparent PETG inspection covers

Implemented hybrid CAD + engineering screening dossier
22 October 2026 | Prototype, not a manufacturing release



10 PETG covers. 398 modeled instances. Metal load paths retained.

The transparent elements are real closed CAD solids with new mounts, not an opacity override on the existing load-bearing arm. Primary frames, heat-spreader cartridges, reducers, bearing seats, brake supports and spring containment stay metal.

Executed: native CAD and mass integration, 17-pose new-part clearance screen, orthotropic cover FEM, fresh baseline/P6 contact simulations, 90 expansion cases and 40 thermal cases. Net mass change: **+53.52 g**.

Open gates: 25/40 thermal cases fail the chosen temperature screen. Print-qualified strength/creep data, mount retention, duty-cycle heat testing and full physical-robot qualification are not available.

02 / Exact conversion map

ID	P5 cover replaced	Outline, mm	Wall, mm	PETG, g
3	J1 rear service cover	diameter 80	4	24.27
33	J2 rear service cover	diameter 80	4	24.27
67	J3 rear service cover	diameter 80	4	24.27
101	J4 rear service cover	diameter 64	3.4	12.94
133	J5 rear service cover	diameter 64	4	15.23
169	J6 rear service cover	diameter 64	4	15.23
210	upper service panel -1	51 x 138	4	34.65
215	upper service panel 1	51 x 138	4	34.65
220	fore service panel -1	51 x 32	4	7.64
225	fore service panel 1	51 x 32	4	7.64

Six rear service caps and four link panels are the only PETG conversions. The J4 cap uses a thinner 3.4 mm wall because a 4 mm cap intersected the forearm. Upper and short panels were enlarged to leave closed mounting-hole ligaments; metal risers prevent the larger panels touching the structural rails.

Unchanged metal: cases 2, 32, 66, 100, 132, 168; frames 272–277; both counterbalance containment assemblies and their spring/reaction parts; all precision drive and gripper load paths. Supplier internals remain catalog-envelope representations.

Every original part has a recorded disposition in assets/part-conversion-map.json. The HTML dossier includes the full searchable 278-part register and all 120 added mount components. Print-local STEP and STL files are supplied for each cover.

03 / Mounts, clearances and retention

Interface	Final modeled dimensions / treatment
PETG holes / sleeve	5.6 mm clearance holes; steel OD 4.0 mm / ID 2.7 mm
Washers / compliant seats	Steel OD 8.0 x 0.7 mm; silicone 0.3 mm installed, 0.5 mm free assumption
Long / short link-panel seat	2.2 mm integral metal riser under the compliant washer
Proposed screws	M2.5 x 10 at caps; M2.5 x 12 at link panels; retained metal threads
Nominal engagement	5.0 mm caps; 5.6 mm J4; 4.8 mm panels; must verify actual thread depth
Load path	Screw -> washer -> metal spacer -> metal land; cover on compliant seat

The floating sleeve is a proposed custom component, not a supplier-rated joint. It avoids direct screw-preload compression of PETG. No heat-set inserts are added to the printed covers and no torque specification is released. Washer capture, gasket compression, retention and seating require physical checks. [3]

Expansion: all 90 assumed cases pass after changing the hole/washer geometry; minimum remaining radial allowance is 0.066 mm. Inputs include 60–100 micrometres/(m K) PETG CTE, 25–50 K rise, 0.20 mm combined positional error and stated size tolerances. These dimensions require inspection/finishing; they are not an as-printed guarantee.

Native geometry: 170 changed/new parts reimport as valid single solids. 1653 candidate new-part intersections at 17 poses found no unintended overlap above 0.05 mm3. This is not continuous sweep, full self-collision or cable/tolerance proof.

04 / Stiffness, stress and numerical limits

3D small-strain orthotropic four-node tetrahedra were solved for four representative covers. The load is 10 N over a 12 mm diameter top patch plus cover self-weight. Finite washer-seat supports are bilateral; nonlinear gasket contact, separation and thread compliance are not solved.

Assumed property	Value
E1 / E2 / E3	1.5 / 1.5 / 0.8 GPa
G12 / G13 / G23	0.54 / 0.28 / 0.28 GPa
nu12 / nu13 / nu23	0.35 / 0.28 / 0.28
Density	1250 kg/m3
Screen thresholds, not certified allowables	12 MPa XY; 5 MPa Z; 4 MPa shear

Cover family	Finest deflection, mm	Last mesh change	Peak screen ratio
Large cap, 4 mm	0.035	8.0%	0.26
J4 cap, 3.4 mm	0.026	9.1%	0.27
Long panel, 4 mm	0.256	9.7%	0.42
Short panel, 4 mm	0.015	7.8%	0.18

Do not read this as a strength certificate. Four mesh sizes, 3.5 to 1.25 mm, were run. The final displacement change remains about 8–10%; results are screening estimates. Support-edge peak stresses are mesh-sensitive. CAD rim fillets are omitted from the extrusion submodels. Several 50 N linear extrapolations exceed the chosen stress screen.

Material inputs are assumed deratings, not a claim that published XY/Z values apply to this print. Suppliers publish different process-specific results, and the exact filament/print orientation still needs coupons. [1,2]

05 / Mass properties and contact dynamics

Rigid body	Mass change, g
bend	1.040
fore	17.701
turret	0.317
fixed	0.317
upper	33.102
roll	1.040
Total	+53.516

Custom-part volume, COM and complete inertia tensors are obtained from BREP integrals and combined by the parallel-axis theorem. Both baseline and P6 use the same retuned reconstructed controller for the comparison below.

P6 10-second contact screen	Computed result
Jaw coordinate range	18.964–38.000 mm; within 17.5–38.5 mm CAD working range
Placement at B / return A	0.139 / 0.134 mm
Peak shoulder command	9.686 N m
Change vs matched metal baseline	+0.096 N m at shoulder
Engine / integration	MuJoCo WASM 3.16.0; 1/2400 s step; 1201 saved states

A real inherited failure was found: the earlier jaw controller overshot to 43.86 mm. Raising simulated velocity damping from 12 to 120 N s/m kept the jaws in range without widening CAD limits or clamping positions. The old failure is retained in the rejected-results folder.

Scope: independent puck contact/friction and rigid-body inertia, not the old 28-scenario electromechanical validation. No coupled cover flexure, full self-collision, rated motor electrical/thermal model or hardware accuracy/safety claim is made.

06 / Temperature, creep and acceptance gates

25 of 40 thermal cases fail the project screen. The clear covers must not be treated as a universally approved replacement for metal caps.

Thermal setup	Screening assumption
Cover model	Two transient wall nodes; conduction, convection and radiation
Transport inputs	k = 0.20 W/(m K); cp = 1200 J/(kg K); emissivity 0.85
Convection / mounts	Inside 3, outside 7.5 W/(m2 K); mount conductance 0.02 / 0.08 W/K
Temperature inputs	Ambient 25 / 40 C; imposed metal case 50–100 C
Chosen cover ceiling	55 C; project screen only, not a PETG continuous-service rating
Example adverse case	Large cap: 25 C ambient / 85 C metal gives about 58–67 C

These inputs are sensitivity assumptions, not measured cooling coefficients. Motor winding temperature, electrical loss and cooling changes caused by replacing a cover are not resolved. Retain a metal cover unless the exact drive/print/mount combination passes heat-soak testing at the intended duty cycle.

Creep: a 1x–8x compliance sensitivity is supplied, not a service-life prediction. At 4x compliance, long-panel deflection under the same 10 N case is 1.026 mm. No calibrated creep hours, fatigue life or permanent-strain value is claimed.

Required tests: print coupons in XY and Z; hole-size/position inspection; sleeve seating and thread engagement; gasket retention, pull-through and vibration; actual-drive thermal cycling; sustained-load creep; repeated service and chemical/UV exposure. PETG covers are not released as fragment containment or safety guards.

07 / Optical proof and render provenance



The new renderer follows rays through the actual closed PETG walls. It includes Fresnel reflection, Snell refraction, total internal reflection, Beer absorption and stochastic scattering. There is no alpha-transparency substitution. [4]

Capture setting	Value / limitation
Resolution and samples	3840 x 2160; 64 camera paths per native pixel; up to 14 interactions
1080p delivery	Geometry-guided spatial denoising; Lanczos downsample in linear radiance
Optical assumptions	IOR 1.57; absorption (3,1.4,0.65)/m; scattering 28/m; HG g = 0.78
Surface appearance	Faint layer/road microtexture; not measured printed clarity
Numerical tests	10 actual C++ optical-function checks pass

Fourteen interactions and finite sampling remain approximation limits. Overlapping dielectric stacks and volumetric next-event sampling are not supported. The final image is an appearance study, not evidence that a particular print will have the same transparency. [5]

08 / Rebuild, provenance and references

Source archive: CAD generators and retained refined BREP inputs; native geometry binaries; full orthotropic FEM/thermal scripts; MuJoCo runtime and two mass-updated models; actual control source and captures; native C++ tracer and numerical tests; STEP/STL outputs and audit JSONs. No font files are distributed.

Run the commands in README.md in order. Engineering reports are generated from the executed output files, and the packaged manifest checks every included file hash. Hash checks are evidence of file integrity, not engineering qualification.

The original refined P5 archive hash is retained in assets/p6-design.json. The source P5 geometry and all unchanged metal geometry are retained. There is no new full motion-video or physically validated disassembly procedure in this delivery.

[1] PolyLite PETG technical data sheet
Primary reference: density 1.25 g/cm3; typical XY/Z modulus 2116.8/1898.7 MPa and tensile strength 50.8/42.8 MPa. These are that supplier's test results, not this print's allowables.
Open primary source

[2] Ultimaker PETG TDS v1.00
Different print-system reference: typical Z modulus 1711 MPa and Z tensile break strength 19.0 MPa. It illustrates why one generic printed-PETG strength is not defensible.
Open primary source

[3] SPIROL compression limiters
Reference for carrying fastener tightening loads through metal rather than crushing plastic. The P6 custom floating sleeve/seat is not a selected or rated SPIROL assembly.
Open primary source

[4] PBRT, fourth edition: Dielectric BSDF
Reference for Fresnel reflection, refraction and radiance transport conventions. P6 uses its own native renderer, not PBRT.
Open primary source

[5] Eastman Spectar Renew 14471-50 copolyester data
A resin-family reference for nD about 1.57, not an optical measurement of the proposed printed covers. The render treats 1.57 as an assumed appearance parameter.
Open primary source

Release status: engineering prototype for review. Printed material calibration, thermal envelope, fastener retention, full swept clearance, EMC/ingress/fire protection and physical robot qualification remain open. Do not operate unguarded hardware based on this dossier.